

D889

Trigonometry

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 12-foot ladder, bought from a hardware store that opened last spring, leans against a school wall and makes a 30 degree angle with the ground. The wall is painted with a school mural. How high up the wall does the ladder reach?

- A. About 6.93 feet
- B. 24 feet
- C. About 10.39 feet
- D. 6 feet

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A teacher uses a unit-circle poster that a former student designed. She asks the class for the exact value of $\sin 210$ degrees. The poster is laminated and hangs by the door. What is the answer?

- A. $1/2$
- B. $-1/2$
- C. square root of $3/2$
- D. -square root of $3/2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A circular garden at a community center has a radius of 10 meters. The garden was planted by volunteers and has a fountain in the middle. A curved path follows the edge for a central angle of 72 degrees. About how long is the path?

- A. About 62.83 meters
- B. About 12.57 meters
- C. About 314.16 meters
- D. 720 meters

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An arc of length 15 cm is on a circle of radius 5 cm. What is the central angle in radians?

- A. 0.33 radians
- B. 75 radians
- C. 3 radians
- D. 3 degrees

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

On a worksheet made for a tutoring session at the public library, $\sin \theta = 3/5$ and θ is in quadrant II. The library session runs 90 minutes. What is $\cos \theta$?

- A. $2/5$
- B. $-3/4$
- C. $4/5$
- D. $-4/5$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

$\cos \theta = 5/13$ and θ is in quadrant IV. What is $\tan \theta$?

- A. $-12/13$
- B. $12/5$
- C. $-12/5$
- D. $-5/12$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

The water depth at a harbor used by a local fishing fleet is modeled by $y = 4 \sin((\pi/6)t) + 10$, with t in hours. The fleet has 14 boats and leaves before dawn. What is the period of the tide?

- A. 12 hours
- B. $\pi/6$ hours
- C. 10 hours
- D. 4 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A Ferris wheel takes 30 seconds per turn. In $y = A \sin(Bt) + D$, what is B ?

- A. $\pi/15$
- B. 30
- C. $15/\pi$
- D. 60π

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Two hiking trails leave a trailhead with a new parking lot. They are 8 km and 5 km long, and the angle between them is 60 degrees. The trailhead has a map made by a scout troop. How far apart are the trail ends?

- A. 7 km
- B. About 9.43 km
- C. It cannot be found with the given data.
- D. 13 km

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A student in an evening class solves $2 \sin x - 1 = 0$ on $[0, 2\pi)$. She uses a graphing calculator borrowed from the math office. Which answer is complete and correct?

- A. $x = \pi/6$ only
- B. $x = \pi/6$ and $x = 5\pi/6$
- C. $x = \pi/6$ and $x = 11\pi/6$
- D. $x = \pi/3$ and $x = 5\pi/3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A drone pilot for a surveying company records a position of $(-3, 3)$ meters from the launch pad. The drone has four rotors and a 20-minute battery. What are the polar coordinates (r, θ) ?

- A. $(6, 3\pi/4)$
- B. $(18, 3\pi/4)$
- C. $(3 \text{ square root } 2, 3\pi/4)$
- D. $(3 \text{ square root } 2, -\pi/4)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A vector has magnitude 10 and direction 150 degrees. What are its components?

- A.** (8.66, 5)
- B.** (-5, 8.66)
- C.** (5, -8.66)
- D.** (about -8.66, 5)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A 12-foot ladder, ~~bought from a hardware store that opened last spring~~, leans against a school wall and makes a 30 degree angle with the ground. ~~The wall is painted with a school mural.~~ How high up the wall does the ladder reach?

Target word: height on the wall **Decides:** Height is opposite the 30 degree angle; ladder is the hypotenuse.

A	Confusion set	Uses tangent, which needs the adjacent side, not the hypotenuse.
B	Inverted operation	Divides by sin 30 instead of multiplying.
C	Confusion set	Uses cosine, which gives the distance along the ground.
D	Correct	$12 \times \sin 30 = 12 \times 0.5 = 6$.

2. Answer B Concept 2

A teacher uses a unit-circle poster ~~that a former student designed~~. She asks the class for the exact value of sin 210 degrees. ~~The poster is laminated and hangs by the door.~~ What is the answer?

Target word: sin 210 **Decides:** Reference angle 30 degrees in quadrant III, where sine is negative.

A	Dropped element	Right size, but drops the negative sign for quadrant III.
B	Correct	$\sin 30 = 1/2$; sine is negative in quadrant III.
C	Confusion set	That is cos 30, with the wrong sign too.
D	Confusion set	That is cos 210 degrees.

3. Answer B Concept 3

A circular garden at a community center has a radius of 10 meters. ~~The garden was planted by volunteers and has a fountain in the middle.~~ A curved path follows the edge for a central angle of 72 degrees. About how long is the path?

Target word: arc length **Decides:** Convert 72 degrees to radians first.

A	Dropped element	That is the whole circumference.
B	Correct	$72 \times \pi/180 = 2\pi/5$; $10 \times 2\pi/5 = 4\pi = \text{about } 12.57$.
C	Adjacent category	That is the garden area (πr^2), not a length.
D	Inverted operation	Multiplies the radius by degrees instead of radians.

4. Answer C Concept 3 • transfer

An arc of length 15 cm is on a circle of radius 5 cm. What is the central angle in radians?

Target word: solve for theta **Decides:** $\theta = s / r$.

A	Inverted operation	Divides r by s.
B	Inverted operation	Multiplies instead of dividing.
C	Correct	$15 / 5 = 3$.
D	Confusion set	s / r gives radians, not degrees.

5. Answer D Concept 4

On a worksheet made for a tutoring session at the public library, $\sin \theta = 3/5$ and θ is in quadrant II. The library session runs 90 minutes. What is $\cos \theta$?

Target word: $\cos \theta$ **Decides:** Size 4/5; negative in quadrant II.

A	Inverted operation	Subtracts $1 - 3/5$ instead of using squares.
B	Confusion set	That is $\tan \theta$.
C	Dropped element	Drops the sign from the quadrant.
D	Correct	$\cos^2 = 1 - 9/25 = 16/25$; cosine is negative in quadrant II.

6. Answer C Concept 4 • transfer

$\cos \theta = 5/13$ and θ is in quadrant IV. What is $\tan \theta$?

Target word: $\tan \theta$ **Decides:** $\sin = -12/13$ in quadrant IV.

A	Confusion set	That is $\sin \theta$, not $\tan \theta$.
B	Dropped element	Drops the negative sign for quadrant IV.
C	Correct	$\sin \theta = -12/13$, so $\tan = (-12/13) / (5/13) = -12/5$.
D	Inverted operation	Flips the ratio: that is $\cos/\sin$.

7. Answer A Concept 5

The water depth at a harbor used by a local fishing fleet is modeled by $y = 4 \sin((\pi/6)t) + 10$, with t in hours. The fleet has 14 boats and leaves before dawn. What is the period of the tide?

Target word: period **Decides:** Period = $2\pi / B$.

A	Correct	$2\pi / (\pi/6) = 12$.
B	Wrong stem number	Reads B as the period.
C	Wrong stem number	That is the midline.
D	Wrong stem number	That is the amplitude.

8. Answer A Concept 5 • transfer

A Ferris wheel takes 30 seconds per turn. In $y = A \sin(Bt) + D$, what is B?

Target word: solve for B **Decides:** $B = 2\pi / \text{period}$.

A	Correct	$2\pi / 30 = \pi/15$.
B	Confusion set	That is the period, not B.
C	Inverted operation	Flips the fraction.
D	Inverted operation	Multiplies 2π by 30 instead of dividing.

9. Answer A Concept 6

Two hiking trails leave a trailhead ~~with a new parking lot. They are 8 km and 5 km long, and the angle between them is 60 degrees. The trailhead has a map made by a scout troop.~~ How far apart are the trail ends?

Target word: third side **Decides:** SAS, so use the law of cosines.

A	Correct	$c^2 = 64 + 25 - 2(8)(5)(0.5) = 49$; $c = 7$.
B	Dropped element	Drops the $-2ab \cos C$ term, using the Pythagorean theorem.
C	Confusion set	SAS is enough with the law of cosines; the law of sines alone would not work.
D	Adjacent category	Adds the two trail lengths.

10. Answer B Concept 7

A student ~~in an evening class~~ solves $2 \sin x - 1 = 0$ on $[0, 2\pi)$. She uses a graphing calculator ~~borrowed from the math office.~~ Which answer is complete and correct?

Target word: all solutions **Decides:** $\sin x = 1/2$ in quadrants I and II.

A	Dropped element	Stops at the calculator answer.
B	Correct	Both quadrants where sine is positive.
C	Confusion set	$11\pi/6$ has sine $-1/2$.
D	Confusion set	Those solve $\cos x = 1/2$.

11. Answer C Concept 8

A drone pilot for a surveying company records a position of $(-3, 3)$ meters from the launch pad. ~~The drone has four rotors and a 20-minute battery.~~ What are the polar coordinates (r, θ) ?

Target word: polar form **Decides:** Point is in quadrant II.

A	Inverted operation	Adds the coordinates instead of using the square root of the sum of squares.
B	Dropped element	Forgets to take the square root.
C	Correct	$r = \text{square root } 18$; angle points into quadrant II.
D	Dropped element	Uses the calculator arctan without fixing the quadrant.

12. Answer D Concept 8 • transfer

A vector has magnitude 10 and direction 150 degrees. What are its components?

Target word: solve for components **Decides:** $x = 10 \cos 150$, $y = 10 \sin 150$.

A	Dropped element	Drops the negative x for quadrant II.
B	Confusion set	Uses the 120 degree values.
C	Inverted operation	Swaps cosine and sine.
D	Correct	$10 \cos 150 = -5$ square root 3; $10 \sin 150 = 5$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	2	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	0	
Confusion set	Every option is real; the test is exact role-matching.	12	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Choosing the right trig ratio	Label the sides relative to the given angle: sine = opposite/hypotenuse, cosine = adjacent/hypotenuse, tangent = opposite/adjacent.	Picking a ratio before labeling sides, or using cosine for the side across from the angle.	A right triangle with one angle, one known side, and one wanted side.
2	Unit circle values and quadrant signs	Use the reference angle for the size of the value and the quadrant for the sign: sine is negative in quadrants III and IV, cosine in II and III.	Using the reference-angle value but forgetting the sign.	An angle beyond 90 degrees, such as 210 degrees or $5\pi/4$.
3	Radians and arc length	Arc length $s = r \times \theta$ works only when θ is in radians; convert degrees by multiplying by $\pi/180$.	Plugging degrees directly into $s = r \times \theta$.	A central angle given in degrees and an arc length asked.
4	Pythagorean identity with quadrant	$\sin^2 + \cos^2 = 1$ gives the size of the missing value; the quadrant gives its sign.	Taking the positive square root every time.	One trig value and a quadrant, asking for another.
5	Sinusoidal models	For $y = A \sin(Bx) + D$, amplitude = $ A $, period = $2\pi/B$, midline $y = D$.	Reading B as the period, or using max value as amplitude.	A periodic situation such as tides or a Ferris wheel, and a request for period, amplitude, or midline.
6	Law of cosines vs. law of sines	Use the law of cosines for SAS or SSS; use the law of sines when you know an angle and its opposite side.	Trying the law of sines with two sides and the included angle, or dropping the $-2ab \cos C$ term.	Two sides and the angle between them.
7	All solutions of a trig equation	A sine or cosine equation usually has two solutions per cycle; find the reference angle, then every quadrant where the sign fits.	Stopping at the calculator's one answer.	Solve on $[0, 2\pi)$ or find all solutions.
8	Rectangular and polar/vector forms	$r = \text{square root of } (x^2 + y^2)$ and θ must point into the correct quadrant; components are $x = r \cos \theta$, $y = r \sin \theta$.	Taking the calculator's $\arctan(y/x)$ without fixing the quadrant.	A point or vector with a negative x value.

Course scope (public description): Trigonometry delves into the core concepts and applications of trigonometric functions and identities, critical for advanced mathematics and practical problem-solving. The course starts with foundational topics, including the definition and calculation of sine, cosine, and tangent ratios, and progresses to solving right triangles and applying trigonometric functions to model real-world periodic phenomena. Students will explore the unit circle, derive fundamental identities, and solve complex trigonometric equations, including applications of vectors, parametric, and polar forms. Emphasizing both theoretical understanding and practical application, this course equips students with the skills necessary for higher-level mathematics courses and various professional fields. No prerequisites are required for this course.